This sample shows a matrix effect. It is also possible to propose a graphical application of the method. An alternative confirmation consists in verifying if relative difference between slope coefficients is included in an acceptance interval, for instance, of ±15% as defined by the European Medicines Agency [10]. If we consider that the line without matrix to be the reference, the relative difference between the two slopes is 33.7%:

$$\frac{|a_1 - b_1|}{a_1} = \frac{|0.0108 - 0.0144|}{0.0108} = 0.337 \text{ or } 33.7\%$$

This value is far from 15% but not surprising because the two lines are obviously non-parallel and it can be concluded that the matrix effect is significant. Consequently, the result must be corrected using the recovery yield to define a correction factor. For this example, the recovery yield is 82.7% and the correction factor is 0.832. For instance, it can be applied to the uncorrected concentration that gives $527 \times 0.832 = 439$. This correction factor can be applied to any unknown sample as far as the matrix is analogous. Result correction is of some consequence on the MU and a full example is presented in Section 8.4.2 showing how to take account of the correction factor in estimating MU.

Resource D Standard addition method (Excel).

	A	B	C	D
1	**Resource D: Standard addition method**			
2		Response		
3	Concentration (ng/ml)	Without matrix	With matrix	
4	0	0.37	6.54	
5	100	1.99	7.64	
6	250	2.95	10.16	
7	500	6.43	13.03	
8	1000	11.60	20.85	
9	1500	16.59	27.99	
10	**Results**			
11	Slope	Intercept		
12	Without matrix			
13	0.0108	0.6382	=LINEST(B4:B9;A4:A9;TRUE;TRUE)	
14	2.805E-04	0.2165		
15	0.997	0.366		
16	1476	4		
17	198	0.536		
18	With matrix			
19	0.0144	6.3250	=LINEST(C4:C9;A4:A9;TRUE;TRUE)	
20	2.368E-04	0.182734888		
21	0.999	0.308959382		
22	3700	4		
23	353	0.382		
24				
25	Uncorrected concentration	527.7	=(B19-B13)/A13	
26	Extrapolated concentration	439.1	=B19/A19	
27	Recovery yield	83.2%	=B26/B25	
28	**Confidence intervals of slopes**			
29	Risk of error	0.05		
30	Student ' s t	2.776		=T.INV(1-B29/2;B16)
31	Without matrix	0.0100	0.0116	=A13-B30*A14
32	With matrix	0.0137	0.0151	=A19-B30*A20

The spiking sample matrix is sometimes misunderstood by analysts, and the main criticism is that the analyte, added as a surrogate calibrant, is not in the same chemical form as the authentic analyte present in the sample. We already have shown in Chapter 1 that in many situations, the compound used for calibration is identical but many solutions exist to get around this problem. May this criticism be valid? Experience shows that, generally, the recovery yield thus obtained is operational and does not impede a good decision.

As already explained, the use of stable isotope labeled (SIL) calibrant or spiking compound may prevent this bias and bring the method closer to a primary method of analysis. This is the recommendation made by official guidelines for the validation of analytical methods for chemical pollutants, such as pesticides, dioxins, polycyclic aromatic hydrocarbon (PAH), or acrylamide. It can be noted that in recent years, the catalogs of commercial reagents have been greatly enriched with this type of molecule.

It has been suggested that when using a single spiked sample, it should have a concentration at least five times that of the test sample, but linearity must be checked before. If the response of the system is nonlinear, the extrapolation involved in the SAM approach becomes more problematic.

There is another extension of SAM in a multidimensional way, called H-point standard addition method (HPSAM), illustrated with an example of drug control in Section 10.1.

2.5 Metrological Approach to Calibration

From a metrological perspective, calibration is defined in International Vocabulary of Metrology (VIM) as:

> "An operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a measurement result from an indication" [11].

This definition of this operation is clearly a two-step procedure as already underlined. If the so-called "relation" is noted as already proposed $Y = f(X) + E$, X is the "measurement standard" value and Y is the "indication". The f function establishes the relationship between X and Y. As discussed in Chapter 1, Y may be diverse: a simple electrical signal, a peak area, a peak height, a ratio between two peaks, etc. This dual nature of calibration may be more precisely established with the mathematical notation:

Step 1. Calibration

$$Y = \alpha_0 + \alpha_1 X + E \tag{2.38}$$

Predicted response for known X

$$Y = a_0 + a_1 X \tag{2.39}$$

Step 2. Inverse predicted concentration

$$Z = \frac{Y - a_0}{a_1} \tag{2.40}$$

The use of Greek letters in Eq. (2.38) and Latin in Eq. (2.39), is aiming to highlight that coefficients a_0 and a_1 are simply estimates of α_0 and α_1, and may vary according to the number of calibrators, their distribution across the calibration range, the number of replicates, the estimation technique, etc. as discussed further. Several approaches are available in the literature to calculate these coefficients, but the most common is OLS regression presented in Section 2.3.1. Because the VIM definition of calibration does not make the difference between the authentic analyte and the calibration compound, which can be authentic or surrogate, some decades ago, several authors proposed to directly fit the reverse function that links X to Y, giving for the linear case:

Step 1. Reverse calibration

$$X = \beta'_0 + \beta'_1 Y + E' \tag{2.41}$$

Step 2. Predicted concentration

$$Z = a'_0 + a'_1 Y \tag{2.42}$$

This approach raises different questions. Originally, the major assumption when using OLS regression is that X is the independent predictive variable known to be without any error and Y is the dependent predicted random variable. In Eq. (2.3), E is the error in Y while the error in X is assumed to be zero or, at least, negligible. Contrarily, in Eq. (2.41), E' is the error in X while there is no error in Y.

Because X and Y do not play the same role in the OLS, as it aims to minimize the sum of squared residuals in the Y-dimension, on the other hand, reverse calibration uses the same criterion, but on X-dimension. Thus, the impact of calibration on the MU is different. There is also a direct link between the values of coefficients a and reverse coefficient a' depending on the coefficient of determination r^2 exhibited by the following equations:

Inverse and reverse calibration links between a and a'coefficients

$$\begin{cases} a'_0 = \dfrac{(1 - r^2)\overline{Y} - a_0}{b_1} \\ a'_1 = \dfrac{r^2}{a_1} \end{cases} \tag{2.43}$$

Equation (2.43) show that both approaches give close results if r^2 is close to 1. This condition is always justified for calibration. A review published some years ago explained the statistical problems raised by both possible approaches [12]. The major drawback is that reverse calibration is relatively easy to apply when the model is a straight line and more difficult for others. While more recent methods use nonlinear calibration models or even require weighed algorithms, reverse calibration did not become popular among analysts except for a few analytical techniques.

The most famous multidimensional application of reverse calibration is near-infrared spectrometry (NIRS), which is widely used to determine proximate components in foods or monitor certain pharmaceutical molecules. In this context, since it is impossible to prepare standard solutions containing pure analytes, calibration is obtained by fitting near-infrared spectra to measurements obtained on the same sample by another reference method. As calibration is performed using full spectra containing many wavelengths it is called *multivariate calibration*. The model becomes:

$$\mathbf{z} = g(\mathbf{Y}) + \mathbf{E}$$

$\mathbf{z}$ is the vector of measurements obtained by the reference method, $\mathbf{Y}$ and $\mathbf{E}$ are matrices – in the mathematical sense – i.e. sets of instrumental responses and errors, traditionally denoted in bold type. This calibration procedure requires specific statistical processing. The best known is the partial least-squares regression (PLS) described in specialized books and out of the scope of this book [13, 14].

2.5.1 Errors in Inverse-predicted Values

As explained in Section 6.6, second step in MU estimation procedure consists in obtaining estimates of the standard deviations of the different identified uncertainty sources. When inverse predicting Z the sample concentration, several sources of uncertainty are included within the instrumental response, such as sample preparation or instrument instability. Whereas Z is a random variable, it is appealing to compute the standard deviation s_Z and use it for MU estimation.

There is some paradox in doing that, while it must be reminded that Z is comparable to X, which is supposed to be error-free to comply with the assumptions of the OLS algorithm. Consequently, Z should also be error-free. Despite this contradiction, an estimate of the standard deviation s_Z of the inverse-predicted concentration was developed and given in Eq. (2.44) [8].

Standard deviation of inverse-predicted concentration Z

$$s_Z = \frac{s_E}{a_1}\sqrt{\frac{1}{J} + \frac{1}{I} + \frac{(Y_k - \overline{Y})^2}{a_1^2 \sum_i (X_i - \overline{X})^2}} \tag{2.44}$$

where:

- s_E is the residual standard deviation of calibration given by Eq. (2.16).
- I is the number of calibrators $1 \le i \le I$.
- J is the number of replicate measurements of the study sample. If no replicate is present, the resulting quotient is $1/J = 1$.
- X_i is the calibrant concentration.
- Y_k is instrument signal of study sample.
- $\overline{X}$ and $\overline{Y}$ are the mean values of concentrations and responses of calibrators, respectively.

This equation is uneasy to use as it requires knowing, in advance, how many replicates will be done. When SAM is used, it is also possible to derive standard deviation

s_{Z^*} of the extrapolated concentration Z^* from Eq. (2.45). This formula will be used for the examples of Section 10.1.

Standard deviation of extrapolated concentration Z^*

$$s_{Z^*} = \frac{s_E}{a_1}\sqrt{\frac{1}{I} + \frac{\overline{Y}^2}{a_1^2 \sum_i (X_i - \overline{X})^2}} \tag{2.45}$$

The use of these standard deviations is not suitable for practical MU estimation and we will propose a more global and comprehensive approach in Section 7.2. But these equations can provide valuable information regarding the optimization of external calibration performances. The following rules inferred from Eq. (2.44) can be applied to better plan the calibration experiment and define some "best calibration practices":

– *Rule 1*. The first equation term includes the residual standard deviation s_E which is an estimate of the random errors of instrumental responses. The possibility of minimizing this parameter is provided by using an appropriate internal standard, as recommended in Section 1.3.
– *Rule 2*. The second term a_1 matches with the slope of the calibration curve. The higher the slope is, the lower the errors in the inverse-computed concentration Z. As explained in [15], a_1 is understandable as the sensitivity of the analytical method, the latter being defined as the ratio between the response variation of the analytical calibration and the analyte quantity variation. An analytical method can thus be considered sensitive when a small variation in the calibrant concentration induces a large variation in the response. This definition differs from international guidelines such Food and Drug Administration (FDA) and European Medicines Agency (EMA), where sensitivity is described as the "lowest measurement range with acceptable accuracy and precision." It must be noted that when the analytical response is corrected by an internal standard, it represents the slope of the ratio and not the authentic analyte response.
– *Rule 3*. At least two strategies for calibrator selection and distribution can be considered to reduce the size of the Z standard deviation and consequently its confidence interval. First, the number I of calibration standards can be increased. According to some guidelines, approximately six calibration points are adequate in many experiments. This point is open to discussion in an integrated approach. The second strategy relies on increasing the number of replicates J, which also reduces the width of the confidence interval.
– *Rule 4*. When the working sample instrumental response Y_k is close to the average point of the calibration range, i.e. approaches mean $\overline{Y}$, the third term inside the square root converges to zero, thus reducing the s_Z value. In the conventional OLS approach, the most precise results are obtained when the measured signal corresponds to a point near the average point (centroid) of the regression line. Obviously, prediction errors are not equal for all points and are smaller when the response is close to the centroid rather than at the edges. This drawback is enhanced when the calibration range is large and response Y becomes heteroscedastic as explained in Section 2.3.2 about WLS regression.